

Music-Operated RGB LED

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Here, RGB LED light intensity and colour change with the rhythm and volume of music. It is a great idea to have such a lighting system at home, gym or music school. This circuit is based on ATmega8A microcontroller (MCU). The author's prototype is shown in Fig. 1.

Circuit and working

The circuit diagram of the music-operated RGB LED is shown in Fig. 2. It is built around electret microphone

(MIC1), 5V voltage regulator 7805 (IC1), op-amp LM358 (IC2), ATmega8A MCU (IC3), RGB LED and a few other components.

Connector CON1 is for connecting 9V power supply. Capacitor C1 is the electrolytic capacitor to reduce low-frequency noise at the input of 7805 (IC1). C3 is the ceramic capacitor to reduce high-frequency noise in the 5V power supply rail.

A pre-amplifier built around op-amp LM358 (IC2) is used to amplify the audio signal received from MIC1. It is configured as a trans-impedance amplifier, and it converts current to voltage level. It is designed to operate with 5V single power supply.

Capacitors C2 and C4 are used to reduce noise in the power supply rails. Resistor R1 is the biasing resistor that sets the current consumption of the microphone. C6 is the coupling capacitor. R1 and C6 together form a high-pass filter.

PARTS LIST	
<i>Semiconductors:</i>	
IC1	- 7805, 5V voltage regulator
IC2	- LM358 operational amplifier
IC3	- ATmega8A MCU
RGB1	- Common-anode RGB LED
<i>Resistors (all 1/4-watt, $\pm 5\%$ carbon):</i>	
R1	- 6.2-kilo-ohm
R2, R7	- 100-kilo-ohm
R3, R6	- 22-kilo-ohm
R4	- 10-kilo-ohm
R5	- 47-ohm
R8, R9	- 120-ohm
R10	- 220-ohm
<i>Capacitors:</i>	
C1	- 100 μ F, 25V electrolytic
C2, C3, C9, C10	- 100nF ceramic disk
C4	- 100 μ F, 16V electrolytic
C5	- 18pF ceramic disk
C6	- 1 μ F, 16V electrolytic
C7, C8	- 10 μ F, 16V electrolytic
<i>Miscellaneous:</i>	
CON1	- 2-pin terminal connector
MIC1	- Electret microphone
	- 28-pin IC base
	- 9V regulated power supply

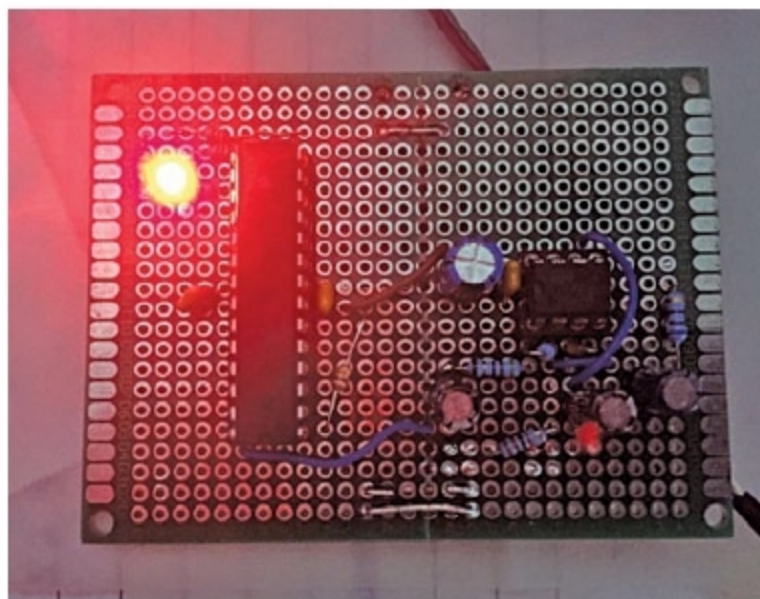


Fig. 1: Author's prototype

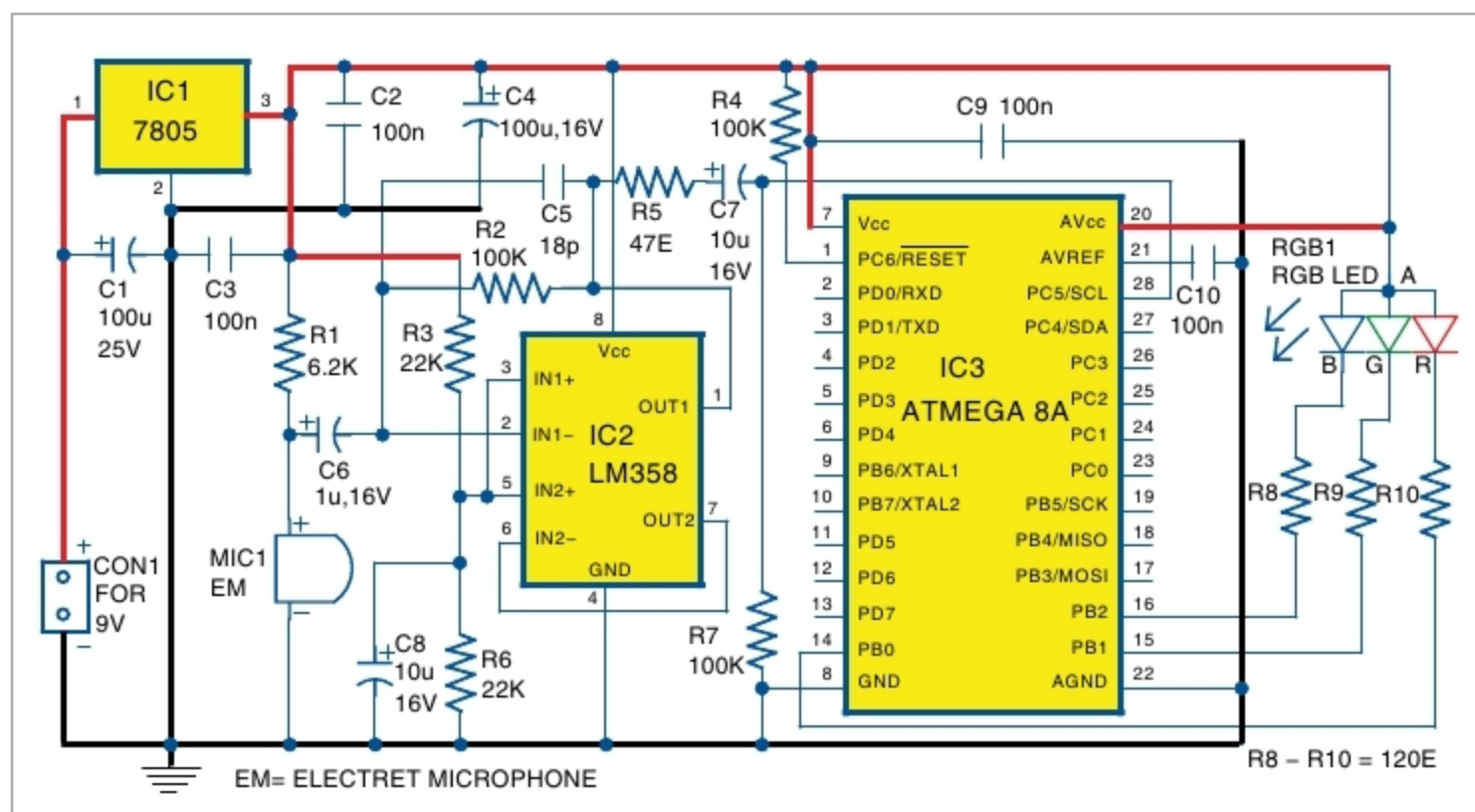


Fig. 2: Circuit diagram of music-operated RGB LED

Resistors R3 and R6 form the voltage divider network for biasing the non-inverting terminal of IC2. R2 is the gain-setting resistor. C5 is the feedback capacitor for stable operation. R7 provides the discharge pathway to prevent the build-up charge on the coupling capacitor (C7).

If charge is allowed to build on C7, it can cause thumps when connected to further stage. Output pin 1 of IC2 is connected to ADC5/PC5 (pin 28) of MCU IC3 through R5 and C7.

IC3 is clocked with an internal RC oscillator and, hence, an external crystal is not used. C9 is the